Week 7: SQL Query Implementation

For the E-Commerce Website, SQL query implementation is used to manage and retrieve data from various database tables such as Customer, Product, Order, Payment, and Review. SQL queries help perform essential operations like inserting records, updating information, deleting data, and generating reports.

1.Display customer name with product name

```
SQL> SELECT C.CUSTOMER_NAME,O.PRODUCT_NAME
  2  FROM CUSTOMER C
  3  INNER JOIN ORDERS O
  4  ON C.CUSTOMER_ID = O.CUSTOMER_ID;

CUSTOMER_NAME              PRODUCT_NAME
-------------------------- ------------------------------
Sweety                     Flower Vase
Arun                       Ceramic Bowl Set
Priya                      Coffee Mug
Kumar                      Ceramic Plate
Meena                      Plant Pot
Ravi                       Wall hanging

6 rows selected.
```

2.Retrieve product names and prices only.

```
SQL> SELECT PRODUCT_NAME,PRICE FROM PRODUCT;

PRODUCT_NAME         PRICE
-------------- ----------
Coffee Cup            125
Flower_Vase           325
Wall hanging          125
Ceramic bowl          100
Plant Pot             150
Ceramic Plate         150

6 rows selected.
```

3.Join order and customer table where customer ID = NULL

```
SQL> SELECT O.*,C.*
  2  FROM CUSTOMER C RIGHT JOIN ORDERS O
  3  ON C.CUSTOMER_ID = O.CUSTOMER_ID
  4  WHERE C.CUSTOMER_ID IS NULL;

no rows selected
```

4.Display customer ID name with product name

```
SQL> SELECT C.CUSTOMER_ID,C.CUSTOMER_NAME,O.PRODUCT_NAME
  2  FROM CUSTOMER C
  3  INNER JOIN ORDERS O
  4  ON C.CUSTOMER_ID = O.CUSTOMER_ID;

CUSTOMER_ID CUSTOMER_NAME             PRODUCT_NAME
----------- ------------------------- ------------------------------
        101 Sweety                    Flower Vase
        102 Arun                      Ceramic Bowl Set
        103 Priya                     Coffee Mug
        104 Kumar                     Ceramic Plate
        105 Meena                     Plant Pot
        106 Ravi                      Wall hanging

6 rows selected.
```

5.Display all orders placed by customers.

```
SQL> SELECT C.CUSTOMER_NAME,O.ORDER_ID,O.TOTAL_AMOUNT,O.ORDER_STATUS
  2  FROM CUSTOMER C JOIN ORDERS O
  3  ON C.CUSTOMER_ID = O.CUSTOMER_ID;

CUSTOMER_NAME              ORDER_ID TOTAL_AMOUNT ORDER_STATUS
------------------------- ---------- ------------ ------------------------
Sweety                          1001          550 Processing
Arun                            1002          250 Dispatched
Priya                           1003          600 Dispatched
Kumar                           1004          200 Dispatched
Meena                           1005         1200 Processing
Ravi                            1006          750 Processing

6 rows selected.
```

6.Find products which price less then ₹300

```
SQL> SELECT PRODUCT_ID,PRODUCT_NAME,PRICE
  2  FROM PRODUCT
  3  WHERE PRICE < 300;

PRODUCT_ID PRODUCT_NAME         PRICE
---------- --------------- ----------
P00002     Coffee Cup             125
P00005     Wall hanging           125
P00001     Ceramic bowl           100
P00003     Plant Pot              150
P00006     Ceramic Plate          150
```

7.Display products available in stock

```
SQL> SELECT * FROM INVENTORY WHERE STOCK_STATUS = 'In Stock';

INVENTORY_ PRODUCT_ID SELLER_ID  STOCK_QUANTITY STOCK_STATUS         LAST_UPDAT
---------- ---------- ---------- -------------- -------------------- ----------
INV001     P00001     S001                120 In Stock             2026-09-20
INV002     P00002     S002                 80 In Stock             2026-09-20
INV003     P00003     S003                 70 In Stock             2026-09-20
INV005     P00005     S005                 50 In Stock             2026-09-20
INV006     P00006     S006                 90 In Stock             2026-09-20
```

8.Low-Stock products

```
SQL> SELECT * FROM INVENTORY WHERE STOCK_QUANTITY< 50;

INVENTORY_ PRODUCT_ID SELLER_ID  STOCK_QUANTITY STOCK_STATUS         LAST_UPDAT
---------- ---------- ---------- -------------- -------------------- ----------
INV004     P00004     S004                20 Low Stock            2026-09-20
```

9.Search products using product name

```
SQL> SELECT * FROM ORDERS WHERE PRODUCT_NAME='Coffee Mug';

  ORDER_ID CUSTOMER_ID ORDER_DAT TOTAL_AMOUNT ORDER_STATUS              CUS
---------- ----------- --------- ------------ ------------------------ ---
CUSTOMER_NAME             PRODUCT_NAME                    QUANTITY
------------------------- ------------------------------ ----------
      1003         103 09-SEP-26          600 Dispatched               YES
Priya                     Coffee Mug                           4
```

10.Display completed orders

```
SQL> SELECT * FROM ORDERS WHERE ORDER_STATUS ='Dispatched';

  ORDER_ID CUSTOMER_ID ORDER_DAT TOTAL_AMOUNT ORDER_STATUS              CUS
---------- ----------- --------- ------------ ------------------------ ---
CUSTOMER_NAME              PRODUCT_NAME                     QUANTITY
------------------------- ------------------------------ ----------
      1002         102 27-AUG-26          250 Dispatched               NO
Arun                       Ceramic Bowl Set                        2

      1003         103 09-SEP-26          600 Dispatched               YES
Priya                      Coffee Mug                              4

      1004         104 15-SEP-26          200 Dispatched               YES
Kumar                      Ceramic Plate                           2
```

11.Display total order count

```
SQL> SELECT COUNT(*) AS TOTAL_ORDER_PLACED  FROM ORDERS WHERE ORDER_DATE BETWEEN '15-AUG-2026' A
ND '15-SEP-2026';

TOTAL_ORDER_PLACED
------------------
                 3
```

12.Orders placed within a specific date range

```
SQL> SELECT * FROM ORDERS WHERE ORDER_DATE BETWEEN '15-AUG-2026' AND '15-SEP-2026';

  ORDER_ID CUSTOMER_ID ORDER_DAT TOTAL_AMOUNT ORDER_STATUS              CUS
---------- ----------- --------- ------------ ------------------------ ---
CUSTOMER_NAME              PRODUCT_NAME                     QUANTITY
------------------------- ------------------------------ ----------
      1002         102 27-AUG-26          250 Dispatched               NO
Arun                       Ceramic Bowl Set                        2

      1003         103 09-SEP-26          600 Dispatched               YES
Priya                      Coffee Mug                              4

      1004         104 15-SEP-26          200 Dispatched               YES
Kumar                      Ceramic Plate                           2
```

13.Display products from lowest to highest price

```
SQL> SELECT * FROM PRODUCT ORDER BY PRICE;

PRODUCT_ID PRODUCT_NAME    CATEGORY_I      PRICE C
---------- --------------- ---------- ---------- -
PRODUCTIMAGE
--------------------------------------------------------------------------------
P00001     Ceramic bowl    00001             100 Y
Elegant white ceramic dinner plate with a smooth finish displayed on a dining ta
ble

P00002     Coffee Cup      00001             125 Y
Beautiful ceramic coffee cup with a simple elegant design

P00005     Wall hanging    00003             125 Y
Decorative ceramic wall hanging with artistic patterns for home decor

PRODUCT_ID PRODUCT_NAME    CATEGORY_I      PRICE C
---------- --------------- ---------- ---------- -
PRODUCTIMAGE
--------------------------------------------------------------------------------

P00003     Plant Pot       00002             150 Y
Decorative ceramic garden planter surrounded by green plants and colorful flower
s outdoors

P00006     Ceramic Plate   00003             150 N
Stylish ceramic dessert plate with a simple modern design for serving sweet dish
es


PRODUCT_ID PRODUCT_NAME    CATEGORY_I      PRICE C
---------- --------------- ---------- ---------- -
PRODUCTIMAGE
--------------------------------------------------------------------------------
P00004     Flower_Vase     00002             325 Y
Decorative ceramic flower vase with an attractive handcrafted design
```

14.Display latest orders first

```
SQL> SELECT * FROM ORDERS ORDER BY ORDER_DATE DESC;

  ORDER_ID CUSTOMER_ID ORDER_DAT TOTAL_AMOUNT ORDER_STATUS              CUS
---------- ----------- --------- ------------ ------------------------ ---
CUSTOMER_NAME              PRODUCT_NAME                     QUANTITY
-------------------------- ------------------------------ ----------
      1006         106 30-SEP-26          750 Processing               YES
Ravi                       Wall hanging                          1

      1005         105 29-SEP-26         1200 Processing               NO
Meena                      Plant Pot                             1

      1001         101 28-SEP-26          550 Processing               YES
Sweety                     Flower Vase                           2


  ORDER_ID CUSTOMER_ID ORDER_DAT TOTAL_AMOUNT ORDER_STATUS              CUS
---------- ----------- --------- ------------ ------------------------ ---
CUSTOMER_NAME              PRODUCT_NAME                     QUANTITY
-------------------------- ------------------------------ ----------
      1004         104 15-SEP-26          200 Dispatched               YES
Kumar                      Ceramic Plate                         2

      1003         103 09-SEP-26          600 Dispatched               YES
Priya                      Coffee Mug                            4

      1002         102 27-AUG-26          250 Dispatched               NO
Arun                       Ceramic Bowl Set                      2


6 rows selected.
```

- **Retirve Unique Values using DISTINCT**
  - Display unique payment category

```
SQL> SELECT DISTINCT PAYMENT_MODE FROM PAYMENT;

PAYMENT_MODE
--------------------
Credit Card
Cash
UPI
Debit Card
Net Banking
```